Second-order behaviour of log phase-type tails and threshold-robust tail index estimation

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[Draft of November 10, 2026]

Abstract

Tail index estimates obtained from the Hill estimator or from a generalised Pareto (GPD) fit to exceedances are notoriously sensitive to the choice of threshold, largely because of the second-order behaviour of the tail that both methods ignore. Building on the eigenvalue plot of Kim and Kim (2015), which fits a two-phase phase-type (PH_2) distribution to log-excesses, we study what this parametric class can and cannot deliver. We give an explicit characterisation of PH_2 with exact parameter ranges: a log-excess is PH_2 if and only if the loss tail is either a signed mixture of two Pareto tails or a mixture of a Pareto and a log-gamma tail with the same index. We show that these two cases satisfy second-order regular variation with $\rho = -\tau < 0$ and $\rho = 0$, respectively, so that log-PH_2 tails cover every second-order parameter $\rho \leq 0$, and we derive the exact threshold bias of the Hill functional under each case. For the Pareto–log-gamma (Exp+Erlang) sub-model we obtain the Fisher information in closed form and show that it becomes singular at the Pareto boundary, where the variance of the tail index estimator grows like $\xi^2/(4p^2)$ in the Erlang weight p. An extensive simulation study ($n = 2000$ and $n = 10\,000$, seven tail models) shows that the Exp+Erlang sub-model gives the most threshold-robust tail index and high-quantile estimates in most settings, that profile-likelihood intervals attain close to nominal coverage wherever the fit is approximately unbiased, and that the Hill estimator remains preferable for tails close to an exact Pareto. In contrast, the two-exponential sub-family is weakly identified, and the sign and size of ρ cannot be recovered from the fit at realistic sample sizes. We give practical recommendations for actuarial tail modelling.

Keywords: Heavy tails; Hill estimator; Phase-type distribution; Second-order regular variation; Threshold selection; Value-at-Risk.
MSC 2020: 62G32; 60E05; 62F12; 62P05.

1 Introduction

Estimating the tail index of a heavy-tailed loss distribution is a routine but delicate task in insurance and finance. Its value determines which moments of the loss exist, and it drives the extrapolation behind high quantiles such as Value-at-Risk (VaR) and tail expectations. The two standard tools are the Hill estimator (Hill, 1975) and maximum likelihood fitting of the generalised Pareto distribution (GPD) to the exceedances over a high threshold, the peaks-over-threshold (POT) method (Balkema and de Haan, 1974; Pickands, 1975; Smith, 1987; McNeil, 1997). Both rely on the first-order approximation of the tail by a Pareto or GPD tail above the threshold, and both suffer from the same well-known difficulty: the estimate depends heavily on the threshold. A low threshold uses

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more data but incurs a bias from the part of the distribution where the first-order approximation is poor; a high threshold reduces this bias at the cost of a large variance (Embrechts et al., 1997; Beirlant et al., 2004; Scarrott and MacDonald, 2012). The size of the bias is governed by the second-order behaviour of the tail, summarised by the second-order parameter $\rho \leq 0$; when ρ is close to zero the bias decays very slowly and no threshold gives a satisfactory trade-off (Drees et al., 2000; de Haan and Ferreira, 2006).

A large literature reduces this sensitivity by modelling the second-order term explicitly, for example through the exponential regression model of Beirlant et al. (1999), the Pareto-departure model of Feuerverger and Hall (1999), or bias-reduced estimators that plug in an external estimate of ρ (Fraga Alves et al., 2003; Gomes and Guillou, 2015). These methods are semi-parametric and target an asymptotic expansion of the tail.

A different, fully parametric route was proposed by Kim and Kim (2015), motivated by the log phase-type (LogPH) class of Ahn et al. (2012). If a loss Y has a regularly varying tail, the log-excess $\log(Y/u)$ over a threshold u has an exponentially decaying distribution, and the Hill estimator is the exponential maximum likelihood estimator (MLE) for these log-excesses. Kim and Kim (2015) replaced the exponential by a phase-type distribution with two phases, PH_2, fitted by the EM algorithm of Asmussen et al. (1996), and plotted the resulting tail index (the eigenvalue of the PH generator closest to zero) against the threshold. Their eigenvalue plot was markedly more stable than the Hill plot across the examples they considered. Two questions were left open: what PH_2 fitting implies for the second-order behaviour of the tail, and whether the gain in stability holds up in a systematic comparison that includes the GPD.

This paper answers both questions, and our conclusions are partly positive and partly negative. Our contributions are as follows.

1. We give an explicit characterisation of PH_2 with exact parameter ranges (Proposition 3.2). A PH_2 law is either a signed mixture of two exponentials, which we call the two-exponential (2E) sub-family, or a mixture of an exponential and an Erlang(2) law with the same rate, the Exp+Erlang (EE) sub-family; each has a Coxian representation, and EE is a boundary limit of 2E. For losses this means that the tail beyond the threshold is either a signed two-Pareto mixture or a Pareto–log-gamma mixture.

2. We show that both log-tails are second-order regularly varying, with $\rho = -\tau$ for 2E and $\rho = 0$ for EE, and we give the auxiliary functions in closed form (Theorem 4.1). Hence log-PH_2 tails cover every $\rho \leq 0$. We derive the exact bias of the Hill functional under each case and contrast it with the threshold invariance of the PH_2 rate parameters (Proposition 4.3).

3. For the EE sub-model, a two-parameter generalisation of the Hill model, we derive the Fisher information in closed form via the exponential integral (Proposition 5.1). We show that it is singular at the Pareto boundary $p = 0$, with $\det I = 4p^2/\mu^2 + O(p^3)$, so that the variance of the tail index estimator grows like $\xi^2/(4p^2)$ (Proposition 5.2). This explains both the cost of the extra flexibility near an exact Pareto tail and the difficulty of identifying ρ, and it motivates profile-likelihood intervals.

4. In a simulation study with seven tail models, two sample sizes and 500 replications, we compare Hill, GPD and several PH_2 variants over a wide range of thresholds (Section 6). The EE sub-model gives the flattest error curves across thresholds and the smallest best-threshold error for most models, while the Hill estimator remains best for tails close to an exact Pareto. The 2E sub-family degrades accuracy when selected, and information criteria do not recover the second-order case.

The paper is organised as follows. Section 2 reviews regular variation, the Hill and POT approaches and phase-type distributions. Section 3 characterises PH_2 and the corresponding log-tails, and Section 4 studies their second-order behaviour. Section 5 develops likelihood inference for the EE sub-model. Section 6 reports the simulation study and Section 7 an empirical illustration. Section 8 gives practical recommendations and discusses limitations. Proofs are collected in Appendix A.

2 Background

2.1 First- and second-order regular variation

Let $Y > 0$ be a loss with distribution function F and tail $\overline{F} = 1 - F$. The tail is regularly varying with index $-\alpha < 0$, written $\overline{F} \in RV_{-\alpha}$, if

$$\lim_{t\to\infty} \frac{\overline{F}(tx)}{\overline{F}(t)} = x^{-\alpha}, \qquad x > 0. \tag{1}$$

Equivalently, $\overline{F}(y) = y^{-\alpha}L(y)$ with L slowly varying. We call α the tail index and write $\xi = 1/\alpha$ for the extreme value index; moments of order less than α are finite.

The rate of convergence in (1) is described by second-order regular variation (Geluk et al., 1997; de Haan and Ferreira, 2006).

Definition 2.1. $\overline{F}$ is second-order regularly varying with first-order index $-\alpha$ and second-order parameter $\rho \le 0$, written $\overline{F} \in 2RV_{-\alpha,\rho}$, if there is a function $A(t) \to 0$ of ultimately constant sign such that

$$\lim_{t\to\infty} \frac{\overline{F}(tx)/\overline{F}(t) - x^{-\alpha}}{A(t)} = x^{-\alpha}\,\frac{x^{\rho}-1}{\rho}, \qquad x > 0, \tag{2}$$

where the right-hand side is read as $x^{-\alpha}\log x$ when $\rho = 0$. Then $|A| \in RV_{\rho}$.

We use the tail-function scale for ρ throughout. If $\overline{F} \in 2RV_{-\alpha,\rho}$, then the tail quantile function $U(t) = F^{\leftarrow}(1 - 1/t)$ is second-order regularly varying with first-order index ξ and second-order parameter ρ/α; the latter is the scale on which ρ is usually reported in the statistics literature (de Haan and Ferreira, 2006). The parameter ρ governs the bias of tail index estimators: for the Hill estimator based on the k largest observations, the asymptotic bias is proportional to $A(n/k)$-type terms, which decay only logarithmically when $\rho = 0$ (Drees et al., 2000; Beirlant et al., 2004).

2.2 The Hill estimator and the POT method

Let $Y_{1:n} \le \cdots \le Y_{n:n}$ be the order statistics of a sample of size n, and let $u = Y_{n-k:n}$ be the threshold leaving k exceedances. Writing $E_j = \log(Y_{n-j+1:n}/u)$, $j = 1, \ldots, k$, for the log-excesses, the Hill estimator is

$$\hat{\xi}^{H}_{k} = \frac{1}{k}\sum_{j=1}^{k} E_j. \tag{3}$$

If $\overline{F}$ is exactly Pareto above u, the log-excesses are i.i.d. exponential with mean ξ, and (3) is the exponential MLE. For high quantiles we use the estimator of Weissman (1978), $\widehat{\mathrm{VaR}}_p = u\{n(1-p)/k\}^{-\hat{\xi}^{H}_{k}}$.

The POT method fits the GPD with shape ξ and scale σ to the excesses $Y - u$ by maximum likelihood (Smith, 1987; McNeil, 1997), and estimates VaR_p by $u + \hat{\sigma}\hat{\xi}^{-1}[\{n(1-p)/k\}^{-\hat{\xi}} - 1]$. Its approximation error above u is governed by the same second-order function A (de Haan and Ferreira, 2006).

2.3 Phase-type and log phase-type distributions

A random variable $X \geq 0$ has a phase-type distribution $\mathrm{PH}(\beta, T)$ with m phases if it is the absorption time of a continuous-time Markov chain with m transient states, initial distribution β (a row vector with $\beta\mathbf{1} = 1$) and sub-generator T (Neuts, 1981; Bladt and Nielsen, 2017). Its survival function and density are

$$\Pr(X > x) = \beta e^{Tx}\mathbf{1}, \qquad f(x) = \beta e^{Tx}t, \quad t = -T\mathbf{1}, \qquad x \geq 0. \tag{4}$$

All eigenvalues of T have negative real parts. If $-\eta_T$ denotes the eigenvalue with the largest real part and β charges the corresponding eigenspace, then $\Pr(X > x)$ decays like $x^j e^{-\eta_T x}$ for some integer $j \geq 0$.

Phase-type laws are closed under excess: for $d \geq 0$,

$$X - d \mid X > d \;\sim\; \mathrm{PH}(\beta_d, T), \qquad \beta_d = \frac{\beta e^{Td}}{\beta e^{Td}\mathbf{1}}. \tag{5}$$

The sub-generator T, and hence η_T, does not depend on d; only the initial vector moves.

Following Ahn et al. (2012) and Kim and Kim (2015), we say that Y has a log phase-type tail above u_0 if $\log(Y/u_0) \mid Y > u_0 \sim \mathrm{PH}(\beta, T)$. By (5), $\log(Y/u) \mid Y > u \sim \mathrm{PH}(\beta_{\log(u/u_0)}, T)$ for every $u \geq u_0$. In particular, $\overline{F} \in RV_{-\eta_T}$ (possibly with a power-of-log factor), so η_T is the tail index whatever the threshold. This threshold stability, the PH analogue of the GPD threshold stability, is the basis of the eigenvalue plot of Kim and Kim (2015): one fits $\mathrm{PH}(\beta, T)$ to the log-excesses over a range of thresholds and plots $1/\hat{\eta}_T$. Note that $m = 1$ gives back the Hill estimator.

3 The two-phase class and its log-tails

With $m = 2$ phases the class is small enough to be described completely, and this description is the key to everything that follows. Dickson and Hipp (2000) observed that a PH_2 law is either a combination of two exponentials or a combination of an exponential and an Erlang(2) law. We make this precise, including the exact parameter ranges.

Lemma 3.1. *The eigenvalues of a 2×2 sub-generator are real and negative.*

Proposition 3.2. *A random variable $X \geq 0$ with no atom at zero is PH_2 distributed if and only if its survival function $S(x) = \Pr(X > x)$ has one of the following two forms:*

(2E) *for some $\theta > 0$, $\tau > 0$ and $-\theta/\tau \leq w \leq 1$,*

$$S(x) = (1-w)e^{-\theta x} + w\, e^{-(\theta+\tau)x}, \qquad f(x) = (1-w)\theta e^{-\theta x} + w(\theta+\tau)e^{-(\theta+\tau)x}; \tag{6}$$

(EE) *for some $\mu > 0$ and $0 \leq p \leq 1$,*

$$S(x) = e^{-\mu x}(1 + p\mu x), \qquad f(x) = (1-p)\mu e^{-\mu x} + p\,\mu^2 x e^{-\mu x}. \tag{7}$$

Case (2E) *arises when T has two distinct eigenvalues $-\theta$ and $-(\theta+\tau)$; case* (EE) *arises when T has a double eigenvalue $-\mu$. Conversely, every law in* (6) *is PH_2 with the Coxian representation*

$$\beta = (1, 0), \qquad T = \begin{pmatrix} -(\theta+\tau) & q(\theta+\tau) \\ 0 & -\theta \end{pmatrix}, \qquad q = \frac{(1-w)\tau}{\theta+\tau} \in [0, 1], \tag{8}$$

and every law in (7) *is PH_2 with $\beta = (1, 0)$ and $T = \begin{pmatrix} -\mu & p\mu \\ 0 & -\mu \end{pmatrix}$.*

The weight w in (6) may be negative: for $-\theta/\tau \le w < 0$ the density is hump-shaped (hypo-exponential), while for $0 < w < 1$ it is a genuine hyperexponential mixture. The tail rate is θ whenever $w < 1$. In the (EE) case the tail rate is μ, and $p = 0$ gives the exponential law of the Hill model.

Remark 3.3 (EE as a boundary of 2E). Put $w = -p\theta/\tau$ in (6). Then $S(x) = e^{-\theta x}\{1 + p\theta(1 - e^{-\tau x})/\tau\} \to e^{-\theta x}(1 + p\theta x)$ as $\tau \downarrow 0$, which is (7) with $\mu = \theta$. The constraint $w \ge -\theta/\tau$ becomes $p \le 1$. Thus the EE sub-family lies on the boundary $\tau = 0$ of the 2E sub-family. Testing EE against 2E is therefore a boundary problem with a nuisance parameter, and standard likelihood-ratio or information-criterion calibrations do not apply (Self and Liang, 1987). We return to this in Section 5.4.

By the threshold stability (5), each sub-family is closed under excess with the same rates, only the weight changing:

$$w_d = \frac{we^{-\tau d}}{1 - w + we^{-\tau d}} \quad \text{(2E)}, \qquad p_d = \frac{p}{1 + p\mu d} \quad \text{(EE)}. \tag{9}$$

Translating to losses gives the two possible log-PH_2 tails.

Corollary 3.4. *Suppose* $\log(Y/u_0) \mid Y > u_0 \sim \mathrm{PH}_2$. *Then for* $y \ge u_0$, *the conditional tail* $\overline{F}_{u_0}(y) = \Pr(Y > y \mid Y > u_0)$ *is either*

$$\overline{F}_{u_0}(y) = (1 - w)\Big(\frac{y}{u_0}\Big)^{-\theta} + w\Big(\frac{y}{u_0}\Big)^{-(\theta+\tau)} \qquad \textit{(signed two-Pareto mixture)}, \tag{10}$$

$$\overline{F}_{u_0}(y) = \Big(\frac{y}{u_0}\Big)^{-\mu}\Big\{1 + p\mu \log\frac{y}{u_0}\Big\} \qquad \textit{(Pareto–log-gamma mixture)}. \tag{11}$$

The law in (11) *mixes, with weights* $1 - p$ *and* p, *a Pareto law with index* μ *and a log-gamma law with shape* 2 *and rate* μ, *both supported on* $[u_0, \infty)$.

4 Second-order behaviour

The two tails in Corollary 3.4 share the first-order index but differ at second order. The second-order condition (2) holds for them exactly, not only asymptotically.

Theorem 4.1. *Let* $\overline{F}$ *have a log-*PH_2 *tail above* u_0, *and write* $s = \log(t/u_0)$ *for* $t \ge u_0$.

(a) In case (EE) *with* $0 < p \le 1$, $\overline{F} \in 2RV_{-\mu,0}$ *with*

$$A(t) = \frac{p\mu}{1 + p\mu s} \sim \frac{1}{\log t}, \qquad \textit{and} \qquad \frac{\overline{F}(tx)}{\overline{F}(t)} - x^{-\mu} = A(t)\, x^{-\mu} \log x \quad \textit{for all } t \ge u_0,\ x \ge 1. \tag{12}$$

(b) In case (2E) *with* $w < 1$ *and* $w \ne 0$, $\overline{F} \in 2RV_{-\theta,-\tau}$ *with*

$$A(t) = \frac{-\tau w e^{-\tau s}}{1 - w + we^{-\tau s}}, \qquad \textit{and} \qquad \frac{\overline{F}(tx)}{\overline{F}(t)} - x^{-\theta} = A(t)\, x^{-\theta}\, \frac{x^{-\tau} - 1}{-\tau} \quad \textit{for all } t \ge u_0,\ x \ge 1. \tag{13}$$

The sign of A *is that of* $-w$, *and* $|A| \in RV_{-\tau}$.

(c) If $p = 0$ *in* (EE) *or* $w = 0$ *in* (2E), *the tail is exactly Pareto above* u_0 *and the left-hand side of* (2) *vanishes identically.*

Corollary 4.2. *The second-order parameters attained by log-*PH_2 *tails are exactly* $\rho \in (-\infty, 0]$*, together with the exact Pareto tail. The value* $\rho = 0$ *is attained only in case* (EE)*, and every* $\rho < 0$ *only in case* (2E) *with* $\tau = -\rho$*.*

Case (2E) belongs to the class of Hall (1982), $\overline{F}(y) = Cy^{-\alpha}\{1 + Dy^{\rho} + o(y^{\rho})\}$, and is the convex (or signed) Pareto mixture considered by Feuerverger and Hall (1999). Case (EE) is the canonical $\rho = 0$ tail: its slowly varying part is $1 + p\mu \log(y/u_0)$, a log factor of the kind that makes the Hill estimator converge slowly.

The next result makes the contrast with the Hill estimator concrete. Let $L(u) = \mathrm{E}[\log(Y/u) \mid Y > u]$ be the population version of the Hill estimator at threshold u.

Proposition 4.3. *Under a log-*PH_2 *tail above* u_0*, the log-excess over any* $u \geq u_0$ *stays in the same sub-family with the same rate parameters, and only the weight changes as in* (9) *with* $d = \log(u/u_0)$*. The Hill functional satisfies:*

(a) in case (EE)*,* $L(u) = \xi(1 + p_d)$ *with* $\xi = 1/\mu$*, so the bias is* $L(u) - \xi = \xi p_d \sim \xi^2 / \log u$*;*

(b) in case (2E)*,* $L(u) = \xi - \xi\, w_d\, \tau/(\theta + \tau)$ *with* $\xi = 1/\theta$*, so the bias is* $O(u^{-\tau}) = O(u^{\rho})$*.*

In contrast, the MLE of μ *(or of* θ*) in the correctly specified* PH_2 *sub-model is consistent for the tail index at every fixed threshold* $u \geq u_0$*.*

Proposition 4.3 formalises the mechanism behind the eigenvalue plot. Under an EE tail, the Hill bias decays only like $1/\log u$, so lowering the threshold to gain data is heavily penalised. The PH_2 fit absorbs the slowly varying factor into the weight p_d, which is re-estimated at each threshold, and the rate parameter is threshold-invariant. This argument is exact only inside the model. Outside it, for example under a log-gamma tail with non-integer shape or a Burr tail, the EE fit can at best absorb the second-order term locally, and its bias must be assessed empirically (Section 6).

5 Likelihood inference for the EE sub-model

5.1 Estimation

Given log-excesses $E_1, \ldots, E_k$ over u, the EE log-likelihood is

$$\ell(\mu, p) = k \log \mu - \mu \sum_{j=1}^{k} E_j + \sum_{j=1}^{k} \log(1 - p + p\mu E_j), \qquad \mu > 0,\ 0 \leq p \leq 1. \tag{14}$$

At $p = 0$ it reduces to the exponential log-likelihood, whose maximiser is the Hill estimator. We maximise (14) directly, using a logistic transform for p, analytic gradients and several starting values. The 2E likelihood is handled in the same way, with $w = -\theta/\tau + (1 + \theta/\tau)\,\mathrm{expit}(c)$ (Appendix B). Direct maximisation of the explicit densities of Proposition 3.2 targets the same estimator as the EM algorithm used by Kim and Kim (2015), is considerably faster, and returns the fitted sub-family. The tail index estimate is $\hat{\xi} = 1/\hat{\mu}$, and a high quantile is estimated by $u \exp\{S^{-1}_{\hat{\mu},\hat{p}}(n(1 - p_{\mathrm{VaR}})/k)\}$, with S from (7).

5.2 Fisher information and its singularity at the Pareto boundary

Let $E_1(a) = \int_a^{\infty} e^{-s} s^{-1}\, ds$ denote the exponential integral (Abramowitz and Stegun, 1964).

Table 1: Asymptotic variance of the Exp+Erlang MLE $\hat{\xi}$, relative to ξ^2 (the asymptotic variance of the Hill estimator under an exact Pareto tail), as a function of the Erlang weight p.

p	0.05	0.10	0.20	0.30	0.50	0.70	0.90
$\sigma^2_\xi(p)/\xi^2$	118.23	33.50	10.07	5.12	2.21	1.25	0.78

Proposition 5.1. *For $0 < p < 1$, put $a = (1-p)/p$ and $J = e^a E_1(a)$. The per-observation Fisher information of the EE model (7) is*

$$I(\mu, p) = \begin{pmatrix} \dfrac{1 + p(1 - a + a^2 J)}{\mu^2} & -\dfrac{1 - aJ}{\mu p} \\ -\dfrac{1 - aJ}{\mu p} & \dfrac{J - p}{p^3} \end{pmatrix}. \tag{15}$$

For $0 < p < 1$ the MLE is asymptotically normal, $\sqrt{k}(\hat{\xi} - \xi) \Rightarrow N(0, \sigma^2_\xi(p))$, with $\sigma^2_\xi(p) = \xi^4 \, [I(\mu, p)^{-1}]_{11}$. The ratio $\sigma^2_\xi(p)/\xi^2$ depends on p only.

We verified (15) against Monte Carlo estimates of the score covariance to three significant digits. Table 1 reports the variance of $\hat{\xi}$ relative to ξ^2, the asymptotic variance of the Hill estimator under an exact Pareto tail. The extra parameter is cheap when the Erlang component is substantial, but becomes very expensive as $p \downarrow 0$.

Proposition 5.2. *As $p \downarrow 0$, $I(\mu, p) \to I_0 = \begin{pmatrix} \mu^{-2} & -\mu^{-1} \\ -\mu^{-1} & 1 \end{pmatrix}$, which is singular. More precisely,* $\det I(\mu, p) = 4p^2/\mu^2 + O(p^3)$ *and*

$$\frac{\sigma^2_\xi(p)}{\xi^2} = \frac{1}{4p^2}\{1 + O(p)\}. \tag{16}$$

At $p = 0$ the two scores are collinear: $\partial_p \ell = -\mu \, \partial_\mu \ell = \mu E - 1$.

At the Pareto boundary, the EE model therefore has a singular information matrix with the true parameter on the edge of the parameter space. Standard $\sqrt{k}$ asymptotics fail there (Self and Liang, 1987; Rotnitzky et al., 2000). To first order, a small Erlang component cannot be told apart from a small change in the tail rate. This is the precise sense in which the departure from an exact Pareto tail, and hence the second-order case, is hard to identify, and it is consistent with the selection behaviour in Section 6.

5.3 Confidence intervals

Wald intervals based on (15) inherit the singularity: when $\hat{p}$ is small they become extremely wide or numerically undefined. We therefore recommend the profile-likelihood interval

$$\Big\{\xi = 1/\mu : \; 2\big[\ell(\hat{\mu}, \hat{p}) - \max_{0 \le p \le 1} \ell(\mu, p)\big] \le \chi^2_{1, 0.95}\Big\}, \tag{17}$$

in which the inner maximisation includes the boundary $p = 0$. Its coverage is examined in Section 6.6.

5.4 The two-exponential sub-family and model selection

The 2E sub-family is less regular still. At $w = 0$ the parameter τ is not identified, and as $\tau \downarrow 0$ with $w\tau$ fixed the model collapses onto EE (Remark 3.3). Selecting between EE and 2E by AIC (Akaike, 1974) or BIC (Schwarz, 1978) is therefore a non-regular comparison: the selection probabilities have no simple relation to the true value of ρ, and fitted values of τ need not estimate $-\rho$. Section 6.4 confirms this empirically.

6 Simulation study

6.1 Design

All data-generating processes (DGPs) have tail index $\alpha = 2$ ($\xi = 0.5$), so that they differ only in their second-order behaviour. They are paired, inside and outside the PH_2 class, at $\rho = 0$ and $\rho = -0.5$:

- PH_2-Erlang ($\rho = 0$, in class): $\log Y \sim 0.5\,\mathrm{Exp}(2) + 0.5\,\mathrm{Erlang}(2, 2)$, i.e. (11) with $\mu = 2$, $p = 0.5$, $u_0 = 1$;
- Loggamma ($\rho = 0$, not in class): $\log Y \sim \mathrm{Gamma}(1.5,\ \text{rate } 2)$;
- Burr ($\rho = -0.5$, not in class): $\overline{F}(y) = (1 + y^{0.5})^{-4}$;
- Two-Pareto ($\rho = -0.5$, in class): $\overline{F}(y) = 0.3y^{-2} + 0.7y^{-2.5}$, $y \geq 1$, i.e. (10) with $\theta = 2$, $\tau = 0.5$, $w = 0.7$;
- GPD ($\rho = -1$): $\xi = 0.5$, $\sigma = 1$, location 0;
- GPD with location 5 ($\rho = -1$; only for $n = 10\,000$): the case for which Kim and Kim (2015) recommend shifting the data by the sample minimum;
- Fréchet ($\rho = -2$): $\overline{F}(y) = 1 - \exp(-y^{-2})$.

For each DGP we draw $R = 500$ samples of size $n = 2000$ (as in Kim and Kim (2015)) and of size $n = 10\,000$. For each sample and each threshold $u = Y_{n-k:n}$ we fit, to the same k exceedances:

- the Hill estimator with the Weissman quantile;
- the GPD MLE on $Y - u$;
- the EE sub-model (E+E);
- the 2E sub-family;
- PH_2 with the sub-family chosen by AIC (AIC) or by BIC.

The threshold grid is $k \in \{25, 50, 100, 150, 200, 300, 400, 600, 800, 1000\}$ for $n = 2000$, and five times these values for $n = 10\,000$, so that k ranges from 1.25% to 50% of the sample. For $n = 10\,000$ we also refit Hill and the PH_2 models to the min-shifted log-excesses $\log\{(Y - m)/(u - m)\}$, $m = \min_i Y_i$ (E+E$_s$); the GPD fit on $Y - u$ is unaffected by the shift. The target quantile is VaR_p with $p = 1 - 1/n$, which requires genuine extrapolation. We report the root mean squared error (RMSE) and bias of $\hat{\xi}$, and the RMSE of $\log(\widehat{\mathrm{VaR}}_p/\mathrm{VaR}_p)$, truncated at ± 3; the truncation is active only for the Hill estimator under the Burr DGP. To summarise sensitivity to the threshold, we report both the best RMSE over the grid (an oracle choice of threshold) and the worst RMSE over the grid, excluding the smallest k. All code, raw estimates and summary files are available with the paper (Section 8).

Table 2: RMSE of $\hat{\xi}$ over the threshold grid $k \in \{50, \ldots, 1000\}$, $n = 2000$, $R = 500$.

		Best k				Worst k			
DGP	ρ	Hill	GPD	E+E	AIC	Hill	GPD	E+E	AIC
PH_2-Erlang	0	0.100	0.073	**0.039**	0.055	0.160	0.702	**0.116**	0.249
Loggamma	0	0.066	0.051	**0.050**	0.185	**0.103**	0.733	0.124	0.259
Burr	-0.5	0.260	0.187	**0.123**	0.147	1.286	0.544	**0.506**	**0.506**
Two-Pareto	-0.5	**0.066**	0.071	0.095	0.219	**0.082**	0.729	0.149	0.285
GPD	-1	0.088	0.044	**0.033**	**0.033**	0.544	0.642	**0.128**	0.278
Fréchet	-2	**0.037**	0.066	0.082	0.081	**0.091**	0.671	0.134	0.239

E+E: PH_2 Exp+Erlang ($\rho = 0$) sub-model; AIC: PH_2 with the sub-family chosen by AIC. "Best" is the smallest RMSE over the threshold grid (oracle threshold); "worst" is the largest RMSE over the grid, a measure of sensitivity to the threshold choice. The smallest value in each row and block is in bold.

6.2 Tail index

Figure 1 plots the RMSE of $\hat{\xi}$ against k for $n = 2000$, and Table 2 summarises it. Three patterns stand out.

First, the GPD MLE is very unstable when the number of exceedances is small: its RMSE at $k = 50$ lies between 0.54 and 0.73 across the DGPs. It improves steadily as k grows and is often the best method at the largest thresholds considered. The Hill estimator behaves in the opposite way. It is accurate at small k, but its bias grows quickly with k unless the tail is close to an exact Pareto; under the GPD and Burr DGPs the bias exceeds 0.5 at $k = 1000$ (Figure 2).

Second, the E+E sub-model has the flattest RMSE curves. Its worst RMSE over the grid is the smallest of all methods for the PH_2-Erlang, Burr and GPD DGPs, and its best RMSE is the smallest for PH_2-Erlang (0.039 against 0.073 for GPD and 0.100 for Hill), Loggamma, Burr and GPD. For the in-class $\rho = 0$ DGP this is what Proposition 4.3 predicts. That the advantage extends to the out-of-class Loggamma and Burr tails shows that the extra log factor absorbs a substantial part of the second-order bias in practice.

Third, the Hill estimator is the best method for the Two-Pareto and Fréchet DGPs. In these tails the second-order term is weak at the thresholds considered, and the Hill estimator's lower variance dominates. E+E pays for its flexibility with a stable negative bias, between -0.12 and -0.05 for $k \geq 50$, consistent with the variance and identifiability cost near $p = 0$ described in Proposition 5.2.

The AIC-selected PH_2 fit is clearly worse than E+E alone for most DGPs; Section 6.4 explains why.

6.3 High quantiles

Table 3 and Figure 3 report the accuracy of $\widehat{\mathrm{VaR}}_{0.9995}$ for $n = 2000$. The differences are smaller than for $\hat{\xi}$, since all methods estimate the quantile from the same exceedances. E+E has a lower quantile RMSE than GPD at every $k \leq 400$ for five of the six DGPs; for Burr it does so only up to $k = 200$ (see `sim/summary_by_k.csv`). The Weissman–Hill quantile is best for the near-Pareto Two-Pareto and Fréchet tails at moderate k, but it deteriorates sharply at large k under GPD and Burr tails. Under the GPD and Burr DGPs all log-scale methods, E+E included, deteriorate once k exceeds roughly 15–30% of the sample: there $\log(Y/u)$ is dominated by observations near zero, outside the tail regime. The GPD fit on the original scale is unaffected by this.

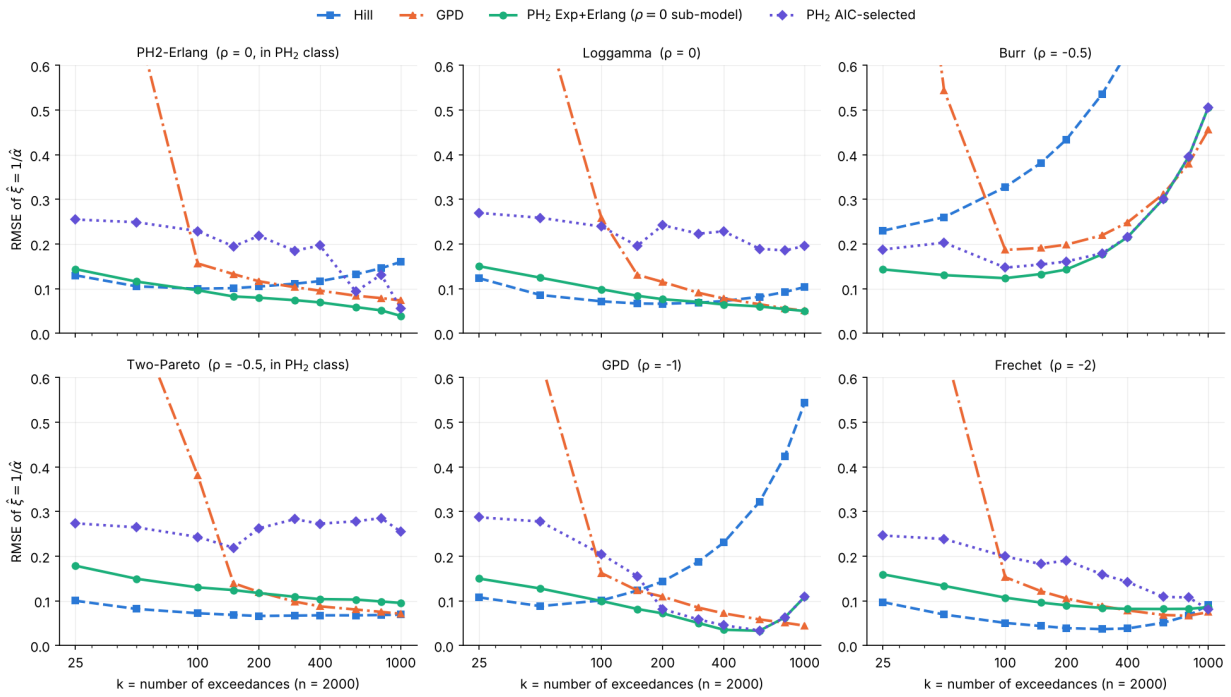


Figure 1: RMSE of $\hat{\xi}$ against the number of exceedances k (log scale), $n = 2000$, $R = 500$.

6.4 Identification of the second-order case

A natural hope, expressed in early versions of this work, is that the fitted PH_2 sub-family indicates which second-order case the data are in: EE for $\rho = 0$ and 2E for $\rho < 0$. The evidence does not support this. Table 4 shows that AIC selects EE in at least half, and usually 70–100%, of the samples for every DGP, including those with $\rho = -0.5$, -1 and -2, while BIC selects EE in more than 90% of samples. Increasing n from 2000 to 10 000 does not change the picture. The only DGP for which selection moves decisively to 2E is the GPD with location 5 fitted without the shift; there the second exponential absorbs the support offset, not a second-order term.

Table 5 shows that selecting 2E is also harmful for estimation. When AIC selects 2E, the RMSE of $\hat{\xi}$ is typically several times larger than when it selects EE, even for the in-class Two-Pareto DGP; the only exception is the Fréchet DGP at the largest threshold. The implied $\hat{\rho} = -\hat{\tau}$ ranges from about -2 to -33, regardless of whether the true ρ is 0, -0.5, -1 or -2. In these fits the faster exponential describes the shape of the log-excess just above the threshold rather than a second-order tail term. These findings agree with the non-regularity described in Section 5.4.

6.5 Larger samples and the min-shift

Table 6 and Figure 4 report the study with $n = 10\,000$, including the min-shifted E+E fit. The conclusions for the unshifted methods are unchanged and sharper: for the PH_2-Erlang DGP the best RMSE of E+E is 0.016, against 0.062 for GPD and 0.081 for Hill.

The min-shift does what it is designed for and no more. For the GPD with location 5, the unshifted Hill and E+E fits have a bias of about -0.2 at large k, while E+E$_s$ attains RMSE 0.019, comparable to the GPD fit (0.021). For DGPs supported on $[0, \infty)$ (GPD, Burr) the shift has no visible effect, so it does not cure the large-k breakdown noted above. For DGPs supported on $[1, \infty)$

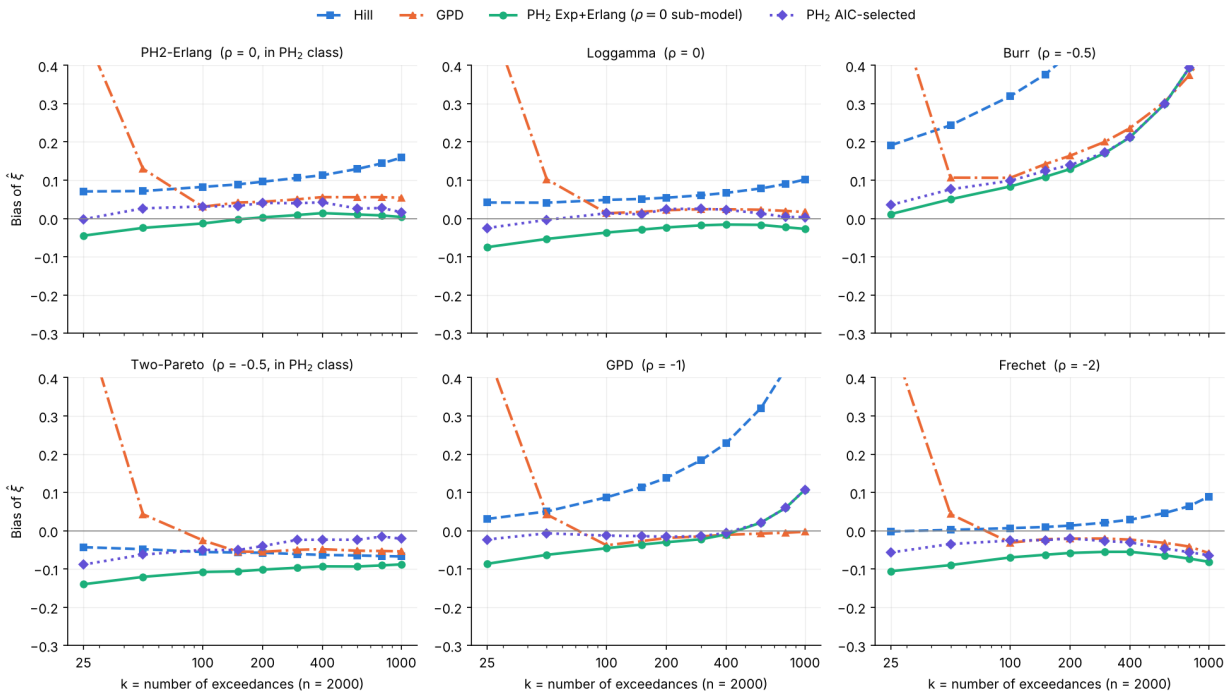


Figure 2: Bias of $\hat{\xi}$ against k, $n = 2000$, $R = 500$.

the shift distorts the fit at large k, since $\log(Y - 1)$ is no longer PH_2; the small best-k RMSE of E+E$_s$ for Loggamma and Two-Pareto in Table 6 occurs where its bias changes sign and does not reflect a uniform improvement. The shift should therefore be applied only when there is evidence that the support starts away from zero, as recommended by Kim and Kim (2015).

6.6 Coverage of interval estimates

Table 7 reports the coverage of nominal 95% intervals for ξ when $n = 2000$. Where the E+E fit is approximately unbiased, the profile-likelihood interval (17) attains close to nominal coverage: 0.92–0.97 under the in-class PH_2-Erlang DGP, 0.88–0.96 under Loggamma, and 0.93–0.96 under the GPD for $k \leq 500$. Wald intervals based on (15) are unreliable, because $\hat{p}$ is close to zero in a substantial fraction of samples (last column), where the interval becomes uninformative. The Hill interval $\hat{\xi}^H \pm 1.96\,\hat{\xi}^H/\sqrt{k}$ under-covers in every setting, often severely, because its bias is large relative to its standard error. No interval can correct a biased point estimate: under the Two-Pareto DGP, where E+E has a bias of about -0.09, and under the GPD at $k = 1000$, where the log-scale fit breaks down, the profile interval also fails. Interval estimates are therefore only as reliable as the threshold region from which they are taken, which reinforces recommendation 1 in Section 8.

7 Empirical illustration

[To be completed by the authors with the Danish fire insurance losses (McNeil, 1997; Resnick, 1997) and/or the exchange-rate data used in Kim and Kim (2015). The script `manuscript/empirical.py` takes a CSV file of losses and produces the threshold plot of $\hat{\xi}$ for Hill, GPD and E+E, with 95% profile-likelihood bands for E+E, together with the AIC sub-family choice and VaR estimates at each threshold. Suggested content: (i) the plot; (ii) the range of k over which E+E is stable,

Table 3: RMSE of $\log(\widehat{\mathrm{VaR}}_{0.9995}/\mathrm{VaR}_{0.9995})$ over $k \in \{50, \ldots, 1000\}$, $n = 2000$, $R = 500$.

		Best k				Worst k			
DGP	ρ	Hill	GPD	E+E	AIC	Hill	GPD	E+E	AIC
PH_2-Erlang	0	0.251	0.229	**0.197**	0.211	0.473	0.434	**0.347**	0.379
Loggamma	0	0.204	0.220	**0.194**	0.229	**0.339**	0.446	0.348	0.385
Burr	−0.5	0.429	0.388	**0.327**	0.351	3.000	**1.193**	2.228	2.228
Two-Pareto	−0.5	**0.137**	0.201	0.189	0.281	**0.268**	0.363	0.294	0.334
GPD	−1	0.286	**0.197**	0.220	0.233	2.551	**0.402**	0.761	0.761
Fréchet	−2	**0.187**	0.237	0.189	0.190	0.465	0.393	**0.320**	0.357

Log-ratio errors are truncated at ±3 before averaging. "Best" is the smallest RMSE over the threshold grid (oracle threshold); "worst" is the largest RMSE over the grid, a measure of sensitivity to the threshold choice. The smallest value in each row and block is in bold.

Table 4: Proportion of samples in which the $\rho = 0$ (Exp+Erlang) sub-family is selected over the two-exponential sub-family ($\rho < 0$).

		$n = 2000$, $k =$						$n = 10\,000$, $k =$					
		AIC			BIC			AIC			BIC		
DGP	ρ	100	400	1000	100	400	1000	500	2000	5000	500	2000	5000
PH_2-Erlang	0	0.79	0.84	0.89	0.97	0.98	0.99	0.78	0.88	0.92	0.99	1.00	1.00
Loggamma	0	0.76	0.79	0.83	0.95	0.98	1.00	0.75	0.82	0.87	0.98	0.99	1.00
Burr	−0.5	0.90	1.00	1.00	0.98	1.00	1.00	0.97	1.00	1.00	1.00	1.00	1.00
Two-Pareto	−0.5	0.72	0.62	0.61	0.94	0.94	0.97	0.60	0.60	0.50	0.95	0.96	0.96
GPD	−1	0.83	0.98	1.00	0.97	1.00	1.00	0.87	0.99	1.00	0.99	1.00	1.00
GPD, $\mu = 5$	−1	–	–	–	–	–	–	0.30	0.00	0.00	0.78	0.01	0.00
Fréchet	−2	0.74	0.79	0.85	0.96	0.98	0.99	0.71	0.82	0.67	0.96	0.99	0.97

The GPD with location $\mu = 5$ was included only in the $n = 10\,000$ study. Fits are on the unshifted log-excesses.

compared with Hill and GPD; (iii) a table of $\hat{\xi}$ and $\widehat{\mathrm{VaR}}_{0.995}$ at a few thresholds; (iv) a remark that the fitted sub-family should not be interpreted as evidence about ρ.]

8 Discussion and recommendations

The two-phase phase-type class fitted to log-excesses has a transparent structure. Its two sub-families correspond exactly to the two second-order regimes, $\rho = 0$ and $\rho < 0$, of a regularly varying tail, and the rate parameters are invariant to the threshold. These properties explain why a PH_2 fit can be much more stable across thresholds than the Hill estimator or a GPD fit.

The simulation evidence, however, points to a more specific conclusion than the one suggested by the theory alone. The useful part of the PH_2 class is the Exp+Erlang sub-model, a two-parameter Pareto–log-gamma tail that generalises the Hill model by a single log factor. It competes with the GPD at the same number of parameters. It tolerates much lower thresholds than the Hill estimator, because its log factor absorbs slowly decaying second-order bias, and much smaller exceedance counts than the GPD, because it is fitted on the log scale where the data are better behaved. The two-exponential sub-family adds little: it is weakly identified, harms accuracy when selected, and does not recover ρ.

For practitioners we suggest the following.

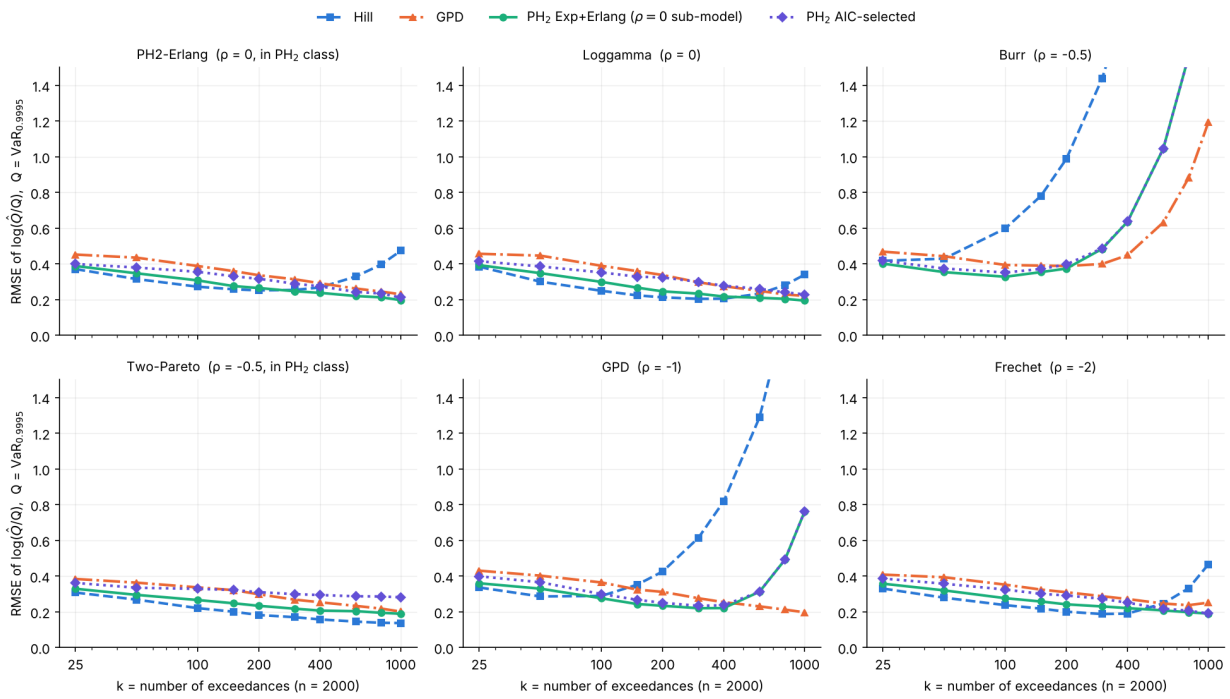


Figure 3: RMSE of $\log(\widehat{\mathrm{VaR}}_{0.9995}/\mathrm{VaR}_{0.9995})$ against k, $n = 2000$, $R = 500$.

1. Plot the E+E tail index estimate against the threshold, with profile-likelihood bands, alongside the Hill and GPD estimates. Choose the tail index from the region in which the E+E estimate is stable. As in Kim and Kim (2015), this region typically begins at a much lower threshold than for the Hill plot.

2. If the E+E fit gives $\hat{p}$ close to zero over the stable region, the data are consistent with a nearly exact Pareto tail. The Hill estimator is then preferable because of its smaller variance (Proposition 5.2).

3. Apply the min-shift of Kim and Kim (2015) only when the support of the losses demonstrably starts away from zero.

4. Do not interpret the PH_2 sub-family selected by AIC or BIC, or a fitted τ, as an estimate of the second-order parameter. If ρ is of interest, use a dedicated estimator (Fraga Alves et al., 2003; Goegebeur et al., 2010).

Several limitations should be noted. Our simulation covers one tail index ($\alpha = 2$) and two sample sizes; tails heavier or lighter than this should be examined. The DGPs supported near zero (GPD, Burr) show that a log-scale method should not be pushed to thresholds in the body of the distribution. Finally, a formal test of $\rho = 0$ against $\rho < 0$ within the PH_2 family would require handling a boundary parameter, an unidentified nuisance parameter and a singular information matrix at once. A parametric bootstrap of the likelihood ratio is a natural candidate, which we leave for future work, together with extensions to inhomogeneous phase-type models (Albrecher and Bladt, 2019).

Table 5: RMSE of $\hat{\xi}$ for the AIC-selected PH_2 fit, split by the selected sub-family, and median $\hat{\rho} = -\hat{\tau}$ among fits selecting the two-exponential sub-family.

n	DGP (ρ)	k	$\#\rho < 0$	$\#\rho = 0$	RMSE, $\rho < 0$	RMSE, $\rho = 0$	median $\hat{\rho}$
2000	PH_2-Erlang (0)	400	78	422	0.473	0.067	-32.72
	PH_2-Erlang (0)	1000	53	447	0.131	0.037	-10.61
	Loggamma (0)	400	104	396	0.484	0.065	-26.35
	Loggamma (0)	1000	84	416	0.465	0.049	-9.27
	Burr (−0.5)	400	1	499	0.388	0.215	-4.24
	Burr (−0.5)	1000	–	500	–	0.506	–
	Two-Pareto (−0.5)	400	191	309	0.418	0.112	-14.03
	Two-Pareto (−0.5)	1000	194	306	0.388	0.103	-8.64
	GPD (−1)	400	12	488	0.188	0.035	-13.22
	GPD (−1)	1000	–	500	–	0.109	–
	Fréchet (−2)	400	104	396	0.265	0.085	-29.95
	Fréchet (−2)	1000	74	426	0.041	0.086	-4.85
10000	PH_2-Erlang (0)	2000	60	440	0.456	0.038	-21.57
	PH_2-Erlang (0)	5000	42	458	0.113	0.016	-2.82
	Loggamma (0)	2000	90	410	0.543	0.043	-20.83
	Loggamma (0)	5000	67	433	0.066	0.035	-3.20
	Burr (−0.5)	2000	–	500	–	0.214	–
	Burr (−0.5)	5000	–	500	–	0.504	–
	Two-Pareto (−0.5)	2000	200	300	0.274	0.088	-3.90
	Two-Pareto (−0.5)	5000	249	251	0.254	0.083	-2.33
	GPD (−1)	2000	4	496	0.120	0.018	-2.06
	GPD (−1)	5000	–	500	–	0.107	–
	GPD, $\mu = 5$ (−1)	2000	500	–	0.077	–	-2.69
	GPD, $\mu = 5$ (−1)	5000	500	–	0.073	–	-3.37
	Fréchet (−2)	2000	89	411	0.319	0.069	-18.57
	Fréchet (−2)	5000	164	336	0.020	0.085	-2.31

Code and data availability

All simulation code, raw per-sample estimates, summary tables and the scripts that produce every table and figure in this paper are available at [repository URL or supplementary material].

Acknowledgements

[To be added.]

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Table 6: RMSE of $\hat{\xi}$ over $k \in \{250, \ldots, 5000\}$, $n = 10\,000$, $R = 500$.

		Best k					Worst k				
DGP	ρ	Hill	GPD	E+E	E+E$_s$	AIC	Hill	GPD	E+E	E+E$_s$	AIC
PH$_2$-Erlang	0	0.081	0.062	**0.016**	0.027	0.035	0.160	0.105	**0.072**	0.124	0.231
Loggamma	0	0.054	0.028	0.034	**0.017**	0.040	0.104	0.100	**0.075**	0.081	0.259
Burr	-0.5	0.242	0.132	**0.086**	**0.086**	0.097	1.282	**0.454**	0.504	0.504	0.504
Two-Pareto	-0.5	0.058	0.057	0.078	**0.013**	0.157	**0.067**	0.105	0.108	0.098	0.206
GPD	-1	0.067	0.020	**0.018**	**0.018**	0.021	0.542	**0.099**	0.107	0.107	0.184
GPD, $\mu = 5$	-1	0.074	0.021	0.096	**0.019**	0.065	0.200	**0.095**	0.200	0.106	0.219
Fréchet	-2	**0.021**	0.038	0.064	0.043	0.067	0.089	0.099	0.085	**0.084**	0.258

E+E$_s$: Exp+Erlang sub-model fitted to the min-shifted log-excesses $\log\{(Y - m)/(u - m)\}$, $m = \min_i Y_i$. For Loggamma and Two-Pareto the small best-k value of E+E$_s$ occurs where its bias changes sign and should not be read as a uniform advantage. E+E: PH$_2$ Exp+Erlang ($\rho = 0$) sub-model; AIC: PH$_2$ with the sub-family chosen by AIC. "Best" is the smallest RMSE over the threshold grid (oracle threshold); "worst" is the largest RMSE over the grid, a measure of sensitivity to the threshold choice. The smallest value in each row and block is in bold.

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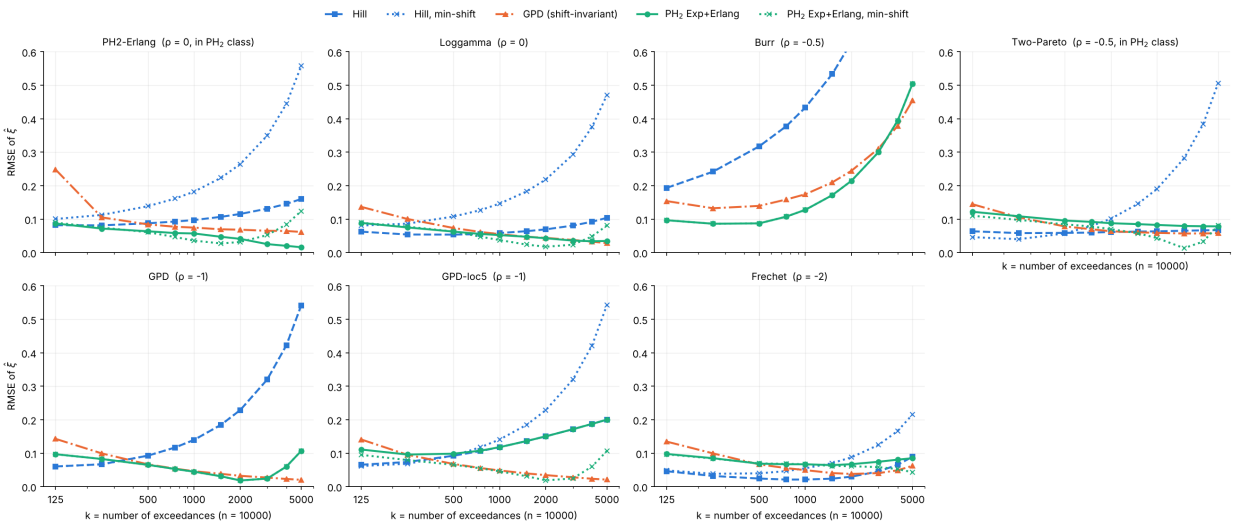


Figure 4: RMSE of $\hat{\xi}$ against k for $n = 10\,000$, with and without the min-shift. For GPD and Burr the shifted and unshifted curves coincide.

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Table 7: Empirical coverage (mean width) of nominal 95% intervals for $\xi = 0.5$, $n = 2000$, $R = 500$.

DGP	k	Wald, E+E	Profile, E+E	Hill	$\Pr(\hat{p} < 0.01)$
PH_2-Erlang	200	0.896 (3.042)	0.966 (0.287)	0.378 (0.165)	0.27
	500	0.920 (1.704)	0.962 (0.227)	0.004 (0.109)	0.15
	1000	0.940 (0.368)	0.924 (0.164)	0.000 (0.082)	0.02
Loggamma	200	0.842 (3.466)	0.964 (0.267)	0.776 (0.153)	0.31
	500	0.850 (2.400)	0.938 (0.212)	0.156 (0.101)	0.22
	1000	0.756 (0.567)	0.878 (0.166)	0.000 (0.075)	0.04
Two-Pareto	200	0.760 (5.510)	0.520 (0.206)	0.498 (0.122)	0.53
	500	0.714 (5.521)	0.126 (0.154)	0.106 (0.076)	0.54
	1000	0.744 (6.119)	0.008 (0.123)	0.006 (0.054)	0.61
GPD	200	0.816 (1.070)	0.930 (0.286)	0.092 (0.177)	0.08
	500	0.976 (0.129)	0.964 (0.136)	0.000 (0.135)	0.00
	1000	0.000 (0.082)	0.000 (0.080)	0.000 (0.129)	0.00

Wald intervals whose width exceeds 10, or whose estimated information is numerically singular (near $\hat{p} = 0$), are treated as uninformative: width 10, counted as covering. Last column: proportion of fits with $\hat{p} < 0.01$.

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A Proofs

Proof of Lemma 3.1. Write $T = \begin{pmatrix} -\lambda_1 & q_1 \\ q_2 & -\lambda_2 \end{pmatrix}$ with $\lambda_i > 0$ and $0 \le q_i \le \lambda_i$. The characteristic polynomial is $\psi^2 + (\lambda_1 + \lambda_2)\psi + \lambda_1\lambda_2 - q_1q_2$, with discriminant $(\lambda_1 - \lambda_2)^2 + 4q_1q_2 \ge 0$, so both roots are real. Their sum $-(\lambda_1 + \lambda_2)$ is negative, and their product $\lambda_1\lambda_2 - q_1q_2$ is positive because absorption is certain, which rules out $q_1 = \lambda_1$ and $q_2 = \lambda_2$ simultaneously. Hence both roots are negative. □

Proof of Proposition 3.2. ($\Rightarrow$) If T has distinct eigenvalues $-\theta > -(\theta + \tau)$, then $e^{Tx} = e^{-\theta x}P_1 + e^{-(\theta+\tau)x}P_2$ for the spectral projections P_1, P_2, so $S(x) = \beta e^{Tx}\mathbf{1} = c_1e^{-\theta x} + c_2e^{-(\theta+\tau)x}$ with $c_1 + c_2 = S(0) = 1$. Put $c_1 = 1 - w$ and $c_2 = w$. Since $f = -S'$ is a density, $f(0) = \theta + w\tau \ge 0$, giving

$w \geq -\theta/\tau$. As $x \to \infty$, $f(x) \sim (1-w)\theta e^{-\theta x}$ unless $w = 1$, so $w \leq 1$. If T has a double eigenvalue $-\mu$, then $T = -\mu I + N$ with $N^2 = 0$, so $e^{Tx} = e^{-\mu x}(I + Nx)$ and $S(x) = e^{-\mu x}(1 + cx)$ with $c = \beta N\mathbf{1}$. Writing $c = p\mu$ gives $f(x) = \mu e^{-\mu x}(1 - p + p\mu x)$; nonnegativity at $x = 0$ gives $p \leq 1$ and nonnegativity as $x \to \infty$ gives $p \geq 0$.

($\Leftarrow$) For (8) the absorption time is $\mathrm{Exp}(\theta+\tau)$ with probability $1-q$ and the sum of independent $\mathrm{Exp}(\theta+\tau)$ and $\mathrm{Exp}(\theta)$ variables with probability q. Its density is

$$(1-q)(\theta+\tau)e^{-(\theta+\tau)x} + q\,\frac{\theta(\theta+\tau)}{\tau}\big(e^{-\theta x} - e^{-(\theta+\tau)x}\big).$$

The coefficient of $\theta e^{-\theta x}$ is $q(\theta+\tau)/\tau = 1 - w$, and that of $(\theta+\tau)e^{-(\theta+\tau)x}$ is $1 - q - q\theta/\tau = w$, as required. Finally, $q \leq 1$ if and only if $w \geq -\theta/\tau$, and $q \geq 0$ if and only if $w \leq 1$. The EE case is the same computation with $\theta + \tau$ and θ both replaced by μ: the absorption time is $\mathrm{Exp}(\mu)$ with probability $1 - p$ and $\mathrm{Erlang}(2, \mu)$ with probability p. □

Proof of Corollary 3.4. $\Pr(Y > y \mid Y > u_0) = S(\log(y/u_0))$ with S from Proposition 3.2. The density of (11) is $(1-p)\mu u_0^{\mu} y^{-\mu-1} + p\mu^2 u_0^{\mu} y^{-\mu-1}\log(y/u_0)$, the stated mixture. □

Proof of Theorem 4.1. (a) With $s = \log(t/u_0)$, $\overline{F}(tx)/\overline{F}(t) = x^{-\mu}\{1 + p\mu(s + \log x)\}/(1 + p\mu s) = x^{-\mu}\{1 + A(t)\log x\}$, which gives (12). Here $A(t) \to 0$ and A is slowly varying, so $\rho = 0$. (b) With $D(t) = 1 - w + we^{-\tau s} > 0$,

$$\frac{\overline{F}(tx)}{\overline{F}(t)} - x^{-\theta} = x^{-\theta}\Big\{\frac{1 - w + we^{-\tau s}x^{-\tau}}{D(t)} - 1\Big\} = x^{-\theta}\frac{we^{-\tau s}(x^{-\tau} - 1)}{D(t)},$$

which equals $A(t)x^{-\theta}(x^{-\tau} - 1)/(-\tau)$ with A as in (13). Since $D(t) \to 1 - w > 0$, $A(t) \sim -\tau w(1 - w)^{-1}(t/u_0)^{-\tau}$, so $|A| \in RV_{-\tau}$ and its sign is that of $-w$. (c) is immediate. □

Proof of Corollary 4.2. By Theorem 4.1, case (EE) with $p > 0$ gives $\rho = 0$, and case (2E) gives $\rho = -\tau$, where $\tau > 0$ is arbitrary; the remaining cases are exactly Pareto. □

Proof of Proposition 4.3. The first statement is (5) applied to the representations of Proposition 3.2, with the weights obtained from $S(d + x)/S(d)$. (a) Under (7) with weight p_d, $\mathrm{E}E = (1-p_d)/\mu + 2p_d/\mu = (1+p_d)/\mu$, and $p_d = p/(1+p\mu d) \sim 1/(\mu \log u)$. (b) Under (6) with weight w_d, $\mathrm{E}E = (1-w_d)/\theta + w_d/(\theta+\tau) = \xi - \xi w_d\tau/(\theta+\tau)$, and $w_d = O(e^{-\tau d}) = O(u^{-\tau})$. Consistency of the MLE at a fixed threshold follows because the log-excesses are i.i.d. from a member of the sub-model with the same rate parameters, and standard MLE theory applies in the interior of the parameter space; at the boundary $p_d = 0$ (exact Pareto) consistency still holds since the Hill estimator is then the restricted MLE. □

Proof of Proposition 5.1. Let $Z = \mu E$ and $D = 1 - p + pZ$. The density of Z is $D\,e^{-z}$, $z > 0$. Differentiating (14) for one observation gives

$$\partial_\mu^2 \ell = -\mu^{-2}(1 + p^2Z^2/D^2), \qquad \partial_\mu\partial_p \ell = \mu^{-1}Z/D^2, \qquad \partial_p^2 \ell = -(Z-1)^2/D^2.$$

Since $D = p(a+Z)$, for any g we have $\mathrm{E}[g(Z)/D^2] = p^{-1}\int_0^\infty g(z)e^{-z}(a+z)^{-1}dz$. Using $\int_0^\infty e^{-z}(a+z)^{-1}dz = J$, $\int_0^\infty ze^{-z}(a+z)^{-1}dz = 1 - aJ$ and $\int_0^\infty z^2e^{-z}(a+z)^{-1}dz = 1 - a + a^2J$, and noting $1 + a = 1/p$, we obtain (15). For $0 < p < 1$ the model is regular, and asymptotic normality of $(\hat{\mu}, \hat{p})$ and the delta method for $\xi = 1/\mu$ give the variance stated. Finally, $[I^{-1}]_{11}\mu^2$ depends on p only, since $I_{11} \propto \mu^{-2}$, $I_{12} \propto \mu^{-1}$ and I_{22} is free of μ. □

Proof of Proposition 5.2. Use the asymptotic series $J = e^a E_1(a) = \sum_{j\geq 0}(-1)^j j!\, a^{-j-1}$ as $a \to \infty$, with $1/a = p/(1-p) = \sum_{i\geq 1} p^i$. Collecting powers of p gives $J = p + p^3 - 2p^4 + 9p^5 + O(p^6)$. Hence

$$\mu^2 I_{11} = 1 + 2p^2 + O(p^3), \qquad \mu I_{12} = -(1 - p + 3p^2) + O(p^3), \qquad I_{22} = 1 - 2p + 9p^2 + O(p^3),$$

and $\mu^2 \det I = (1+2p^2)(1-2p+9p^2) - (1-p+3p^2)^2 + O(p^3) = 4p^2 + O(p^3)$. Then $\sigma_\xi^2/\xi^2 = \mu^2 [I^{-1}]_{11} = I_{22}/(\mu^2 \det I) = \{4p^2\}^{-1}\{1 + O(p)\}$. The scores at $p = 0$ are $\partial_\mu \ell = \mu^{-1} - E$ and $\partial_p \ell = \mu E - 1$, which are collinear. We confirmed the expansion numerically in 60-digit arithmetic. □

B Computational details

For the 2E sub-family we use $\theta = e^a$, $\tau = e^b$ and $w = -\theta/\tau + (1 + \theta/\tau)\,\mathrm{expit}(c)$, which maps $\mathbb{R}^3$ onto the full parameter range of Proposition 3.2. The log-likelihood is $-\theta \sum_j E_j + \sum_j \log\{(1 - w)\theta + w(\theta + \tau)e^{-\tau E_j}\}$. For EE we use $\mu = e^a$ and $p = \mathrm{expit}(c)$. Both likelihoods are maximised by L-BFGS-B with analytic gradients from 18 (2E) and 9 (EE) starting values centred at the Hill estimate. The GPD is fitted by Nelder–Mead from three starting values. One replication with ten thresholds takes about one second at $n = 2000$ on a single core.